Skittles Lab

[ECON 4753] — *University of Arkansas*

Raw Data

Each bar represents one pack, with the colored sections showing how many Skittles of each color it contains. This plot shows the variation from pack to pack and makes it possible to compare the total counts and color composition of individual packs. Are the packs similar in their color composition, or are some colors noticeably more common in particular packs? The plot also gives us a first look at whether the packs appear to contain a consistent number of Skittles.

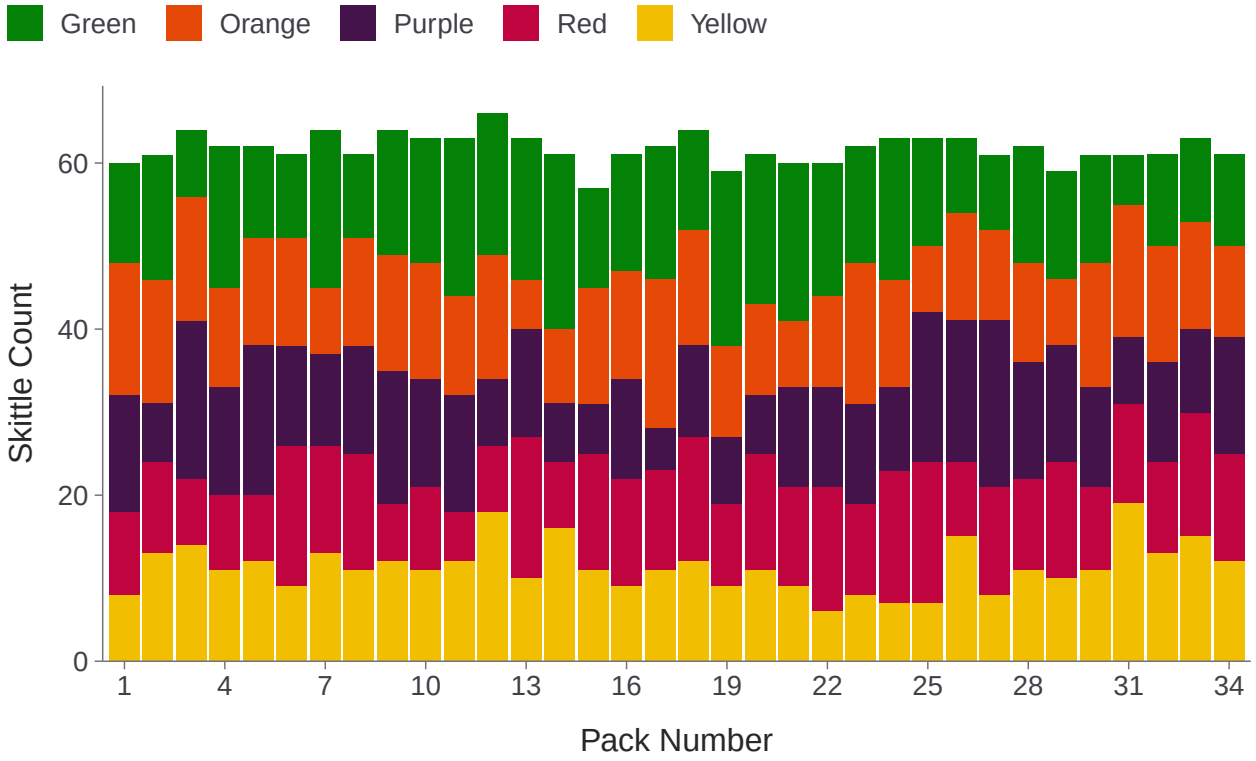


Figure 1 — Color Counts by Pack

Number of Skittles per Pack

The average number of Skittles per pack is 61.74 with a standard deviation of 1.76. Is this variation small enough to describe the packs as consistent, or do pack totals differ substantially? The histogram shows how the total number of Skittles is distributed across the packs collected by the class. The solid curve is the smoothed density of the observed totals, while the dashed curve is a

normal distribution with the same mean and standard deviation.

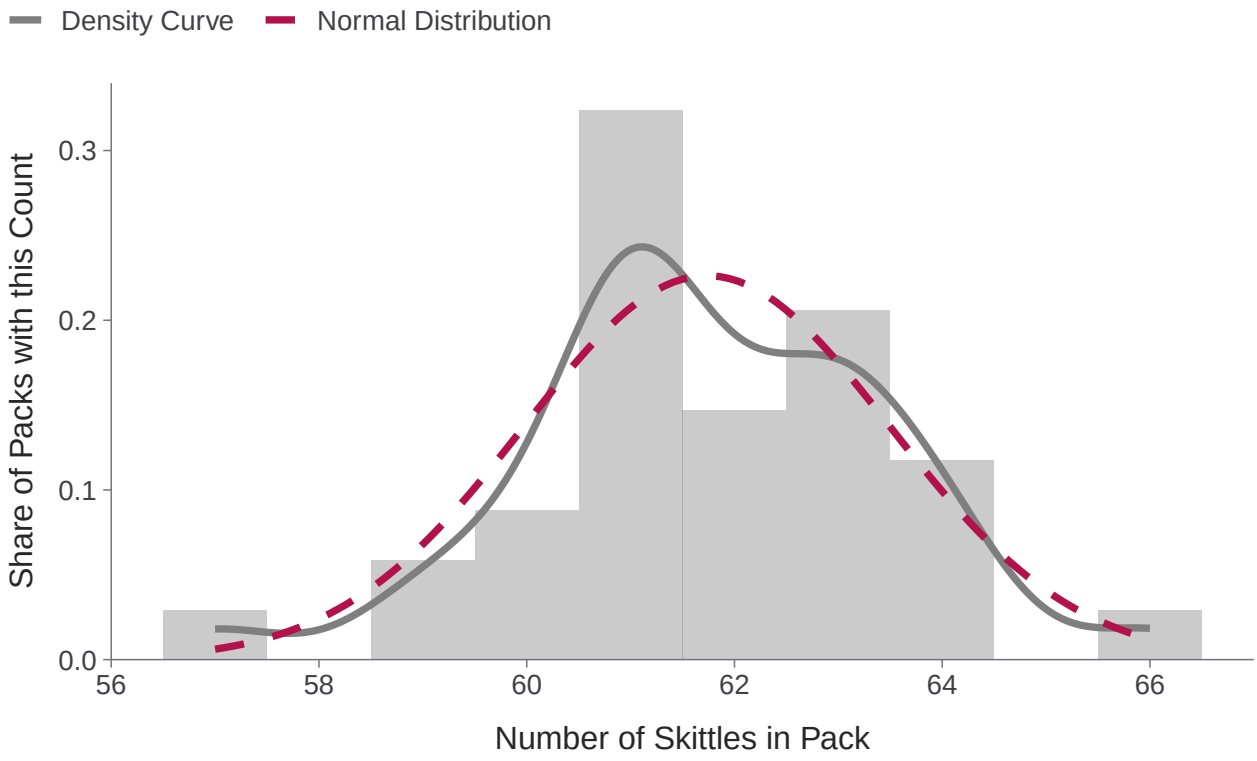


Figure 2 — Total Number Skittles in a Pack

Color Proportions in Early Draws

This figure compares the color proportions observed after the first one, two, three, and four draws. Comparing the groups shows how the observed mix changes as more Skittles are included in the sample. With only a few draws, how much can the observed color mix vary just by chance? Does the mix become more representative of the available colors as the number of draws increases?

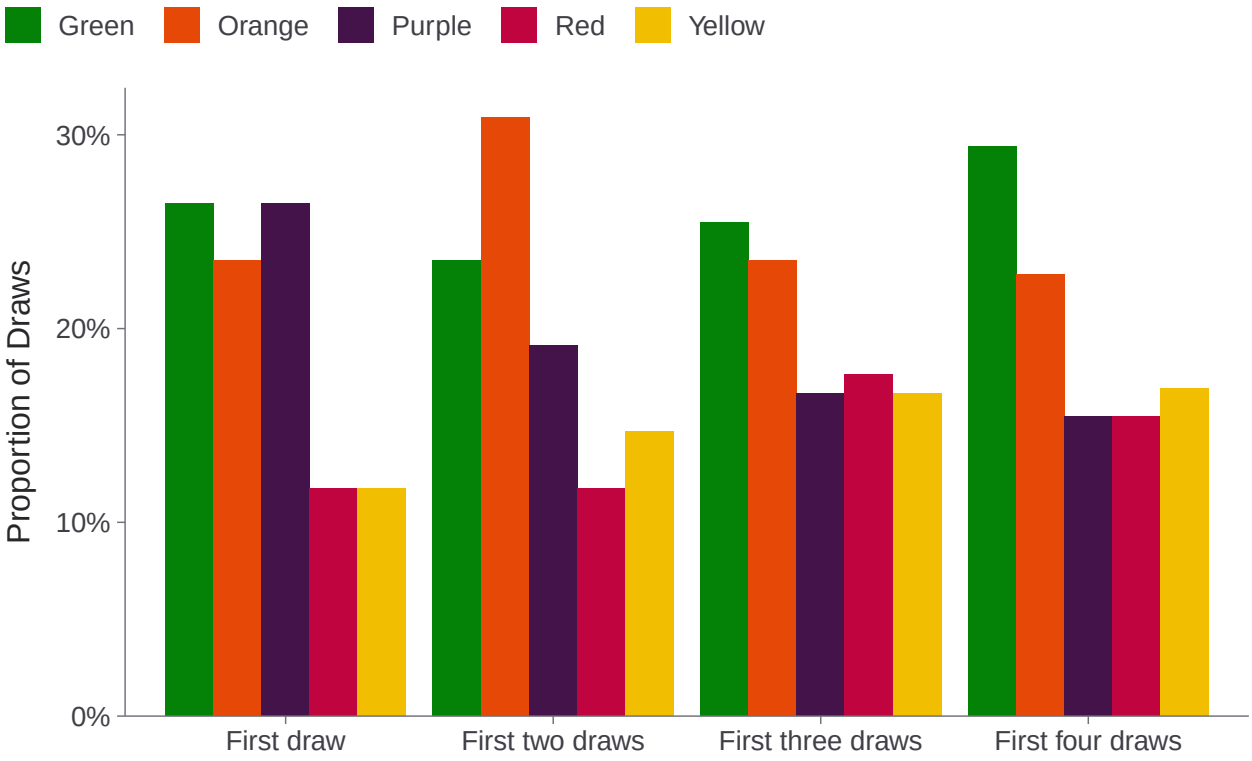


Figure 3 — Color Proportions in the First Three Draws

Distribution of Each Color

The five panels show the distribution of each color's share within a pack. The horizontal position of a pack depends on the proportion of its Skittles that have that color, so the panels can be compared for center and spread. The dashed vertical line at 20% marks the expected share when all five colors are equally common. Is the average share of each color close to 20%? Which colors are most consistent from pack to pack, and which show the greatest variation?

Table 1: Summary of Sample Distribution

Color	Average Proportion in Pack	Std. Dev. of Proportion across Packs
Red	0.20	0.04
Orange	0.20	0.04
Yellow	0.18	0.04
Green	0.22	0.06
Purple	0.20	0.06

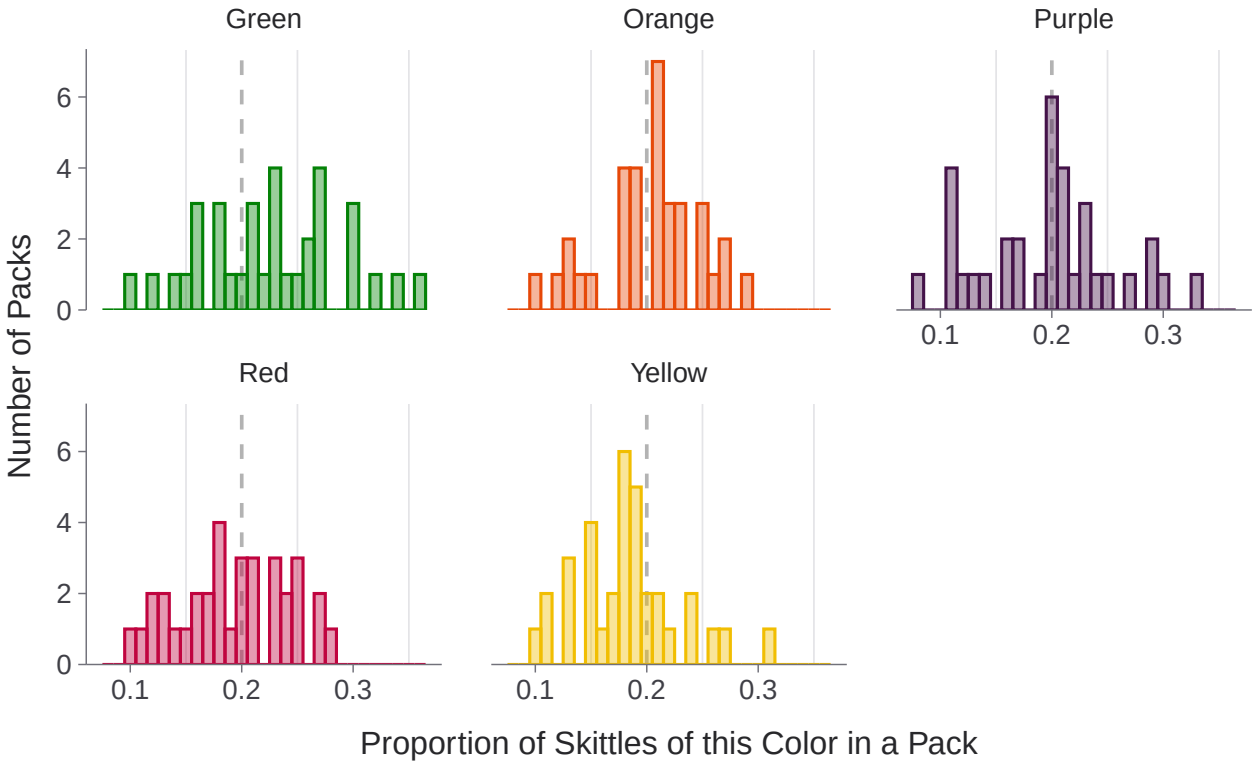


Figure 4 — Distribution across Repeated Sampling

What is the average share of each color per pack?